Following standard terminology from quantum optics, we refer to all such combinations of $\hat{x}$ and $\hat{p}$ as *quadrature operators*. The rotated quadratures obey the same commutation relations as $\hat{x}$ and $\hat{p}$,

$$[\hat{x}(\theta), \hat{p}(\theta)] = i. \tag{3.55}$$

In addition to rotations, translations of the quadratures are also important. These are realised by the *displacement operator*, defined as

$$\hat{D}(\alpha) = e^{\alpha\hat{a}^\dagger - \alpha^*\hat{a}}. \tag{3.56}$$

Since the exponent is anti-Hermitian, the displacement operator is unitary. We have $\hat{D}^\dagger(\alpha) = \hat{D}(-\alpha)$. Furthermore, using the Campbell identity again[3]

$$\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha) = \hat{a} + \alpha. \tag{3.57}$$

From here, we can see that the displacement shifts $\hat{x}$ and $\hat{p}$

$$\hat{D}^\dagger(\alpha)\hat{x}\hat{D}(\alpha) = \hat{x} + \sqrt{2}\,\mathrm{Re}(\alpha), \tag{3.58}$$

$$\hat{D}^\dagger(\alpha)\hat{p}\hat{D}(\alpha) = \hat{p} + \sqrt{2}\,\mathrm{Im}(\alpha). \tag{3.59}$$

This means that for any (normalised) pure state $|\psi\rangle$, a displacement $|\psi\rangle \to \hat{D}(\alpha)|\psi\rangle$ shifts the expectation values of $\hat{x}$ and $\hat{p}$ by $\sqrt{2}\,\mathrm{Re}(\alpha)$ and $\sqrt{2}\,\mathrm{Im}(\alpha)$, respectively. The variances, on the other hand, do not change (see Prob. 3.3). The operator $\hat{D}(\alpha)$ thus shifts the mean of states but leaves their noise properties unchanged (at least in $\hat{x}$ and $\hat{p}$). In this sense, it is a displacement. Below, we will also see that this corresponds to translations in phase space.

3.1.5 Coherent states

In addition to the energy eigenstates, there are of course many other interesting classes of quantum states of harmonical oscillators. One particularly important class is that of so-called *coherent states*. Coherent states are ubiquitous in quantum optical experiments and applications because they provide an accurate description of the state of light produced by lasers. Hence, they are among the easiest states to produce in practice. At the same time, they are in a sense the 'most classical' states of the quantum harmonic oscillator.

Formally, the coherent states are the eigenstates of the annihilation operator. For every complex number α, there exists a state $|\alpha\rangle$ such that

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle. \tag{3.60}$$

That is, $|\alpha\rangle$ is a left-eigenstate of $\hat{a}$. It is also a right-eigenstate of $\hat{a}^\dagger$ with eigenvalue α^*, i.e. $\langle\alpha|\hat{a}^\dagger = \langle\alpha|\alpha^*$. We take $|\alpha\rangle$ to be normalised. The mean number of excitations is seen to be

$$\langle\alpha|\hat{n}|\alpha\rangle = |\alpha|^2. \tag{3.61}$$

[3] For operators $\hat{A}$ and $\hat{B}$ such that $\hat{A}$ commutes with the commutator $[\hat{A}, \hat{B}]$, the Campbell identity simplifies to $e^{\hat{A}}\hat{B}e^{-\hat{A}} = B + [\hat{A}, \hat{B}]$. We can apply this result here with $\hat{A} = -(\alpha\hat{a}^\dagger - \alpha^*\hat{a})$ and $\hat{B} = \hat{a}$.

For coherent states of light, this corresponds to the mean photon number. Using (3.20) and (3.21), it is also easy to find the expectation values

$$\langle\alpha|\hat{x}|\alpha\rangle = \sqrt{2}\,\mathrm{Re}(\alpha), \tag{3.62}$$

$$\langle\alpha|\hat{p}|\alpha\rangle = \sqrt{2}\,\mathrm{Im}(\alpha). \tag{3.63}$$

How about a general expression for the state? Since $\hat{a}$ and $\hat{a}^\dagger$ destroy and create excitations, clearly $|\alpha\rangle$ cannot have support on any restricted set of Fock states but must contain contributions from all possible numbers of excitations. Expressing $|\alpha\rangle$ in the Fock basis

$$|\alpha\rangle = \sum_{n=0}^{\infty} c_n|n\rangle, \tag{3.64}$$

and applying $\hat{a}$, we get

$$\hat{a}|\alpha\rangle = \sum_{n=1}^{\infty} \sqrt{n}c_n|n-1\rangle = \alpha\sum_{n=0}^{\infty} c_n|n\rangle = \alpha\sum_{n=1}^{\infty} c_{n-1}|n-1\rangle, \tag{3.65}$$

which implies that $c_n = \frac{\alpha}{\sqrt{n}}c_{n-1} = \ldots = \frac{\alpha^n}{\sqrt{n!}}c_0$. The last coefficient c_0 is fixed by normalisation. We then get the Fock-state basis expansion

$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}|n\rangle. \tag{3.66}$$

Note that for $\alpha = 0$, the coherent state equals the $n = 0$ Fock state, i.e. the ground state. In general, the number of excitations in a coherent state follows a Poisson distribution with parameter $|\alpha|^2$

$$p(n) = |\langle n|\alpha\rangle|^2 = \frac{|\alpha|^{2n}e^{-|\alpha|^2}}{n!}. \tag{3.67}$$

This is, for example, the distribution of the number of photons that would be detected by an ideal photon counter measuring a coherent state of light.

What about the distributions of $\hat{x}$ and $\hat{p}$? Using (3.42) and properties of the Hermite polynomials, one can also find the $\hat{x}$-space wavefunction $\psi_\alpha(x) = \langle x|\alpha\rangle$ of a coherent state

$$\psi_\alpha(x) = \frac{1}{\pi^{1/4}}e^{-\frac{1}{2}|\alpha|^2}e^{-(x-\alpha/\sqrt{2})^2+x^2/2}. \tag{3.68}$$

This can also be expressed in terms of the expectation values of $\hat{x}$ and $\hat{p}$, which is sometimes more convenient or intuitive. Defining $x_\alpha = \sqrt{2}\,\mathrm{Re}(\alpha)$ and $p_\alpha = \sqrt{2}\,\mathrm{Im}(\alpha)$, we have

$$\psi_\alpha(x) = \frac{1}{\pi^{1/4}}e^{ip_\alpha(x-\frac{1}{2}x_\alpha)}e^{-\frac{1}{2}(x-x_\alpha)^2}. \tag{3.69}$$

From here, we see that the distribution of x is Gaussian

$$|\psi_\alpha(x)|^2 = \frac{1}{\sqrt{\pi}}e^{-(x-x_\alpha)^2}, \tag{3.70}$$

centered at x_α with variance $\frac{1}{2}$. Similarly, computing the $\hat{p}$-space wavefunction, the distribution of p is found to be Gaussian with the same variance centered at p_α. In fact, the variance is $\frac{1}{2}$ for any rotated quadrature $\hat{x}(\theta)$. In particular, the vacuum state corresponds to Gaussian distributions centered at zero.

It is no coincidence that all coherent states display the same variances for $\hat{x}$ and $\hat{p}$ as the vacuum. In fact, coherent states can be understood as displaced vacuum states. From (3.57), it follows that the state $\hat{D}(\alpha)|0\rangle$ is an eigenstate of $\hat{a}$ with eigenvalue α

$$\hat{a}\hat{D}(\alpha)|0\rangle = \hat{D}(\alpha)\hat{D}^\dagger(\alpha)\hat{a}\hat{D}(\alpha)|0\rangle = \hat{D}(\alpha)(\hat{a}+\alpha)|0\rangle = \alpha\hat{D}(\alpha)|0\rangle. \tag{3.71}$$

The state is also normalised since $\hat{D}(\alpha)$ is unitary. Hence

$$|\alpha\rangle = \hat{D}(\alpha)|0\rangle. \tag{3.72}$$

Since there are coherent states centered at any x and p and with any mean excitation number, it is perhaps intuitive that they should span the entire Hilbert space. This is indeed the case. However, unlike the Fock states, distinct coherent states are not orthogonal. The overlap of two coherent states is

$$\langle\beta|\alpha\rangle = e^{i\,\mathrm{Im}(\beta^*\alpha)}e^{-\frac{1}{2}|\alpha-\beta|^2} = e^{-\frac{1}{2}|\alpha|^2-\frac{1}{2}|\beta|^2+\beta^*\alpha}, \tag{3.73}$$

which is always nonzero (but decreases exponentially as α and β move apart). The coherent states form an over-complete basis and it can be shown that

$$\frac{1}{\pi}\int |\alpha\rangle\langle\alpha|\, d^2\alpha = \mathbb{1}, \tag{3.74}$$

where the integral is over the complex plane, $d^2\alpha = d\,\mathrm{Re}(\alpha)d\,\mathrm{Im}(\alpha)$.

In what sense are coherent states the 'most classical' states, as claimed at the beginning of this subsection? This statement is based on their time dynamics. Coherent states keep their shape under the time evolution generated by the harmonic-oscillator Hamiltonian (they evolve without dispersion) and their mean values for $\hat{x}$ and $\hat{p}$ behave exactly like the position and momentum of a classical one-dimensional harmonic oscillator. At the same time, they are minimum-uncertainty states that saturate the uncertainty relation for $\hat{x}$ and $\hat{p}$. So they are stable states with 'position' and 'momentum' that evolve like those of a classical oscillator while adding the minimum amount of noise which quantum mechanics allows.

To see that coherent states evolve like classical oscillators, first note that

$$e^{-i\theta\hat{n}}|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2}\sum_{n=0}^{\infty}\frac{\alpha^n e^{-i\theta n}}{\sqrt{n!}}|n\rangle = |e^{-i\theta}\alpha\rangle. \tag{3.75}$$

The time evolution under the Hamiltonian (3.22) starting from a coherent state $|\psi(0)\rangle = |\alpha\rangle$ is then

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\alpha\rangle = e^{-i\frac{\omega}{2}t}|e^{-i\omega t}\alpha\rangle. \tag{3.76}$$

The evolved state is thus still a coherent state at all times (the global phase can be dropped) and the expectation values of $\hat{x}$ and $\hat{p}$ evolve as

$$\langle\hat{x}(t)\rangle = \sqrt{2}\,\mathrm{Re}(e^{-i\omega t}\alpha) = \cos(\omega t)\langle\hat{x}(0)\rangle + \sin(\omega t)\langle\hat{p}(0)\rangle, \tag{3.77}$$

$$\langle\hat{p}(t)\rangle = \sqrt{2}\,\mathrm{Im}(e^{-i\omega t}\alpha) = \cos(\omega t)\langle\hat{p}(0)\rangle - \sin(\omega t)\langle\hat{x}(0)\rangle, \tag{3.78}$$

which is exactly the expected behaviour for a classical oscillator (with rescaled x and p).

To see that coherent states are minimum-uncertainty states, first note that the Robertson uncertainty relation (2.24) for $\hat{x}$ and $\hat{p}$ gives

$$\langle \Delta^2 \hat{x} \rangle \langle \Delta^2 \hat{p} \rangle \geq \frac{1}{4} |\langle [\hat{x}, \hat{p}] \rangle|^2 = \frac{1}{4}, \tag{3.79}$$

for any state. In any coherent state, $\langle \Delta^2 \hat{x} \rangle = \frac{1}{2} = \langle \Delta^2 \hat{p} \rangle$, as can be seen from the Gaussian wavefunction or by direct calculation. Coherent states therefore saturate the uncertainty relation, with the uncertainty distributed evenly between $\hat{x}$ and $\hat{p}$.

3.1.6 Phase space

In classical physics, the *phase space* of a given system is the set of all possible states of the system in some chosen parametrisation. For example, the state of a mechanical system of particles can be described by fixing all the position and momentum of each particle. Each point in phase space corresponds uniquely to one state of the system. Phase-space pictures (sometimes called phase-space 'portraits') are very useful tools for illustrating and understanding the behaviour of physical systems.

In quantum mechanics, because conjugate variables do not commute, there is not a one-to-one correspondance between states and points in phase space. E.g. no state of the quantum harmonic oscillator can have well-defined values of $\hat{x}$ and $\hat{p}$ simultaneously. Nevertheless, the concept of phase space and phase-space pictures are still extremely useful, in particular for harmonic oscillators.

Fig. 3.2 illustrates the phase space of a harmonic oscillator. We can extend the classical phase-space picture to the quantum case by including an illustration of the uncertainty in the variables. This gives an intuitive depiction of quantum states of the oscillator. Moreover, it can be made rigorous by means of the Wigner function, which is a distribution over phase space, as we will now discuss.

3.1.7 Phase-space distributions

Because $\hat{x}$ and $\hat{p}$ are incompatible observables, it is not generally possible to define a joint probability distribution $p(x, p)$ over phase space for quantum states. However, it is possible to define phase-space distributions that retain some of the features of probability distributions while at the same time capturing all the same information as a given density operator $\hat{\rho}$. Such *quasiprobability distributions* thus provide alternative descriptions of quantum states, and all the information which can be computed from $\hat{\rho}$, such as expectation values, distributions of measurement outcomes etc., can also be derived from them (at least in principle – depending on the quantity in question, one or the other description may be more convenient for computing it). In fact, it is possible to construct a consistent phase-space formulation of the entire theory of quantum mechanics, including operators and measurements. We will not delve deeply into the details of that here, but in the following we present some useful quasiprobability distributions.